

Voice-Controlled OBSTACLE-AVOIDING ROBOT Using Bluetooth And Arduino Uno

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The smart rover is a versatile robotic platform equipped with Bluetooth and voice control capabilities. The obstacle avoidance mechanism, facilitated by sensors, enables the smart rover to navigate autonomously, avoiding collisions with objects in its path.

The smart rover presents a promising solution for robotics enthusiasts and educators seeking

an accessible and engaging platform for exploring concepts of robotics, automation,

BILL OF MATERIALS

Components	Description	Quantity
Arduino Uno (MOD1)	Microcontroller	1
Robot chassis	4-wheel drive chassis	1
Geared DC motors (M1-M4)	5V-6V DC motors	4
Servo motor (SM1)	Micro geared servo	1
Ultrasonic sensor (SEN1)	HC-SR04	1
Motor driver shield (MOD2)	L329D	1
Bluetooth module HC-05	Bluetooth module	1
Battery	5V DC	1
Switch (S1)	Switch	1
Jumper wires	For connection	30

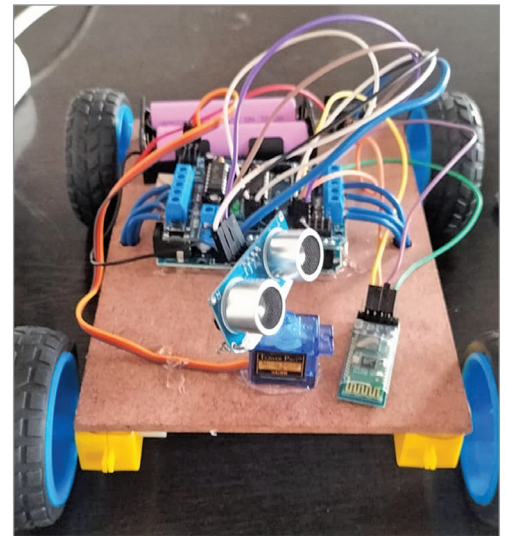


Fig. 1: Authors' prototype

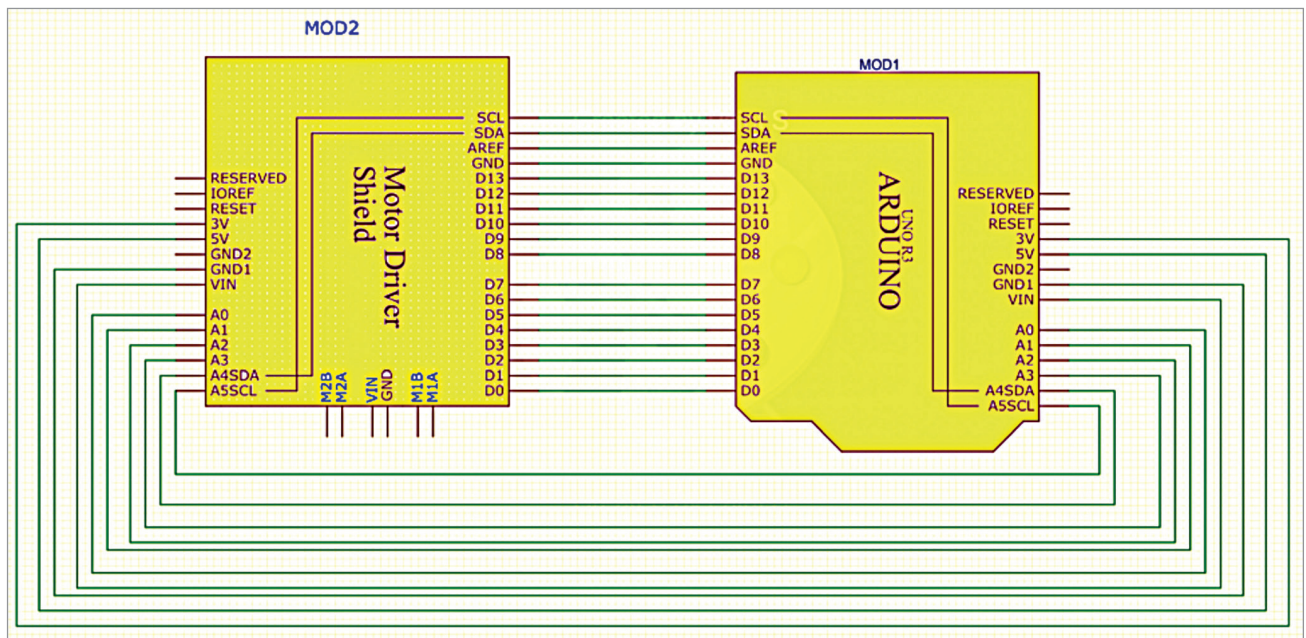


Fig. 2: Circuit diagram of motor driver shield with Arduino Uno